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REVIEW

Crops and genotypes differ in efficiency of potassium uptake and use

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Cultivars with increased efficiency of uptake and utilization of soil nutrients are likely to have positive environmental effects through reduced usage of chemicals in agriculture. This review assesses the available literature on differential uptake and utilization efficiency of K in farming systems. Large areas of agricultural land in the world are deficient in K (e.g. 3/4 of paddy soils in China, 2/3 of the wheatbelt in Southern Australia), with export in agricultural produce (especially hay) and leaching (especially in sandy soils) contributing to lowering of K content in the soil. The capacity of a genotype to grow and yield well in soils low in available K is K efficiency. Genotypic differences in efficiency of K uptake and utilization have been reported for all major economically important plants. The K-efficient phenotype is a complex one comprising a mixture of uptake and utilization efficiency mechanisms. Differential exudation of organic compounds to facilitate release of non-exchangeable K is one of the mechanisms of differential K uptake efficiency. Genotypes efficient in K uptake may have a larger surface area of contact between roots and soil and increased uptake at the root–soil interface to maintain a larger diffusive gradient towards roots. Better translocation of K into different organs, greater capacity to maintain cytosolic K^+ concentration within optimal ranges and increased capacity to substitute Na^+ for K^+ are the main mechanisms underlying K utilization efficiency. Further breeding for increased K efficiency will be dependent on identification of suitable markers and compounding of efficiency mechanisms into locally adapted germplasm.

Introduction

Farming systems have become increasingly efficient over the years by (1) lowering labour inputs because of mechanization and (2) increasing yields via improved cultivars and increased fertilizer inputs. However, increased fertilizer demand because of growing global population and the expansion of agriculture into less-productive areas has increased the cost of fertilizers and decreased fertilizer-use efficiency. As an example, nitrogen-use efficiency in cereal production (wheat, maize, rice, barley, sorghum, millet, oats and rye) worldwide is approximately 33%, with the unaccounted 67% representing a US$15.9 billion annual loss of N fertilizer (assuming fertilizer–soil equilibrium) (Raun and Johnson 1999).

Various agronomic measures can be employed to increase efficiency of fertilizer use, such as better matching nutrient supply to crop demand by using slow-release fertilizers (Wiesler 1998), optimizing fertilizer placement and planting geometry (Yadav et al. 1997), employing precision agriculture systems (Wong et al. 2005), etc. Growing genotypes that have increased

Abbreviation – QTL, quantitative trait loci.

efficiency of nutrient uptake and use is a more recent attempt to increase fertilizer-use efficiency (Rengel and Marschner 2005, Trehan 2005).

Efficiency of nutrient uptake and use

Nutrient deficiencies in plants are widespread throughout much of the agricultural regions in the world, especially in developing countries where fertilizers may not be affordable (Sillanpää and Vlek 1985, Snapp et al. 2002, Welch et al., 1991). Nutrient deficiencies may be the result of an inherently low nutrient status of the soil, low mobility of nutrients within the soil or poor solubility of the given chemical form of the nutrient. The solubility and mobility of nutrients within the soil is governed by a number of factors, including mass flow of water, adsorption capacity of the soil, soil pH (Jungk 2001, Shuman 1991), the chemical form of nutrients (Crowley and Rengel 1999) and plant-induced changes in the soil immediately surrounding roots (=rhizosphere) (Rengel and Marschner 2005).

In contrast to developing countries, in developed countries plentiful amounts of fertilizers are applied, resulting in abundant inorganic nutrients in the system that may cause disturbance in normal biogeochemical cycles of nutrients (Drinkwater and Snapp 2007, Kayser and Isselstein 2005). However, this tendency is being challenged by the current emphasis on developing more efficient cultivars for sustainable, low-input agriculture, fuelled by increasing cost of fertilizers, government restrictions on fertilizer applications to minimize environmental damage and the increased use of poor-quality land (Lynch 1998, Rengel 2005, Rengel and Marschner 2005, Trehan 2005).

Some plant species (Brennan and Bolland 2007, El-Dessougi et al. 2002, Rose et al. 2007, Trehan and Claassen 1998) and genotypes within species (Damon et al. 2007, Damon and Rengel 2007, Gunes et al. 2006, Marschner et al. 2007, Valizadeh et al. 2003, Yang et al. 2004) have a capacity to grow and yield well on soils low in available nutrient; these species and genotypes are considered tolerant to nutrient deficiency (=nutrient efficient) (Rengel 2005, Rengel and Marschner 2005). Efficient genotypes have specific physiological mechanisms allowing them to gain access to sufficient quantities of a specific nutrient (uptake efficiency) and/or to more effectively utilize amounts of that nutrient taken up (utilization efficiency) (Sattelmacher et al. 1994). Using nutrient-efficient genotypes in combination with soil fertilization is the optimal nutrient management strategy for resilient and sustainable farming systems.

Nutrient-efficient genotypes are important in modern agriculture because they can produce greater yields on soils where the effectiveness of fertilizers may be limited by chemical and biological reactions, topsoil drying, subsoil constraints and/or disease interactions (Graham and Rengel 1993, Lynch 1998, Rengel and Marschner 2005). Growing nutrient-efficient genotypes of crop plants on soils of low nutrient availability represents an environmentally friendly approach that would reduce land degradation by (1) reducing the use of machinery (Thongbai et al. 1993) and (2) minimizing application of chemicals on agricultural land.

Breeding new nutrient-efficient genotypes adapted to low nutrient environments has so far been sporadic. For macronutrient-efficient genotypes, breeding has advanced to a considerable extent only for P, and to a much smaller extent for N (see Rengel 2005) and K (Stelling et al. 1996, Trehan 2005, Wu et al. 1998). Proportionally more was done on micronutrient-efficient genotypes (e.g. Fe, Zn and Mn; Rengel and Marschner 2005). Zn- and Mn-efficient barley and wheat varieties are routinely grown in many parts of wheatbelt in southern Australia.

Potassium in farming systems

Global consumption of K has increased at an average rate of 4.4% per annum over the period 1999–2005 (IFA 2005). Consumption is predicted to increase by a further 12% to 35.2 Mt K_2O by 2010 (Heffer and Prud'homme 2006). Using rice as an example (4–8 tonnes of grain per ha result in exporting about 56–112 kg of K from the soil; Yang et al. 2004), the annual K demand for irrigated rice by 2025 is estimated to be 9–15 Mt, which is equivalent to about 1.7 times the K requirement in 1990 (Dobermann et al. 1998). Although global K reserves are likely to be sufficient for hundreds or even thousands of years of agriculture (Sheldrick 1985), the profitability of agricultural production on marginal soils will rely increasingly on the efficient use of K and other nutrients.

Historically, the use of K fertilizers has been limited in many countries with extensive agricultural production (including Australia), with crops and pastures relying on native soil reserves of K. Presently, about one quarter of the arable soils and three quarters of paddy soils in China are K deficient for rice (see Yang et al. 2004). Long-term field experiments across five Asian countries indicated a negative K balance in irrigated rice systems; e.g. in Mekong River delta in Vietnam, such negative balances can be as high as -86 kg K ha^{-1} $year^{-1}$ (Nguyen My et al. 2006), with removal of straw having a particularly severe impact on K budget. In south-western Australia, the incidence of K deficiency in wheat has increased steadily, with two thirds of the arable soils prone to K depletion through continued removal in hay or grain and straw (Pal et al. 2001b). Leaching of K, especially in sandy soils (Kayser and Isselstein 2005), is also a significant contributor to poor K-use efficiency in farming systems.

Root systems of annual crops generally occupy <1% of the soil volume (cf. Barber 1995), allowing roots to get in direct contact with only a tiny proportion of soil K. Therefore, K uptake by plants is highly dependent on the soil capacity to deliver K to the root surface by mass flow and diffusion, the latter being the predominant process (Jungk 2001, Jungk and Claassen 1997, Liebersbach et al. 2004). The rates of diffusion are influenced by the plant capacity to take up K at the root surface and create a diffusive gradient from the bulk soil towards the root surface. A reader is referred to recent review papers addressing the plant structures and processes associated with K uptake by roots (Ashley et al. 2006, Gambale and Uozumi 2006, Gierth and Maser 2007). Other aspects of the K role in plant metabolism are dealt with elsewhere (e.g. in alleviation of abiotic stresses; Cakmak 2005).

The present review will focus on the physiological and genotypic differences in K efficiency, and how these differences can be exploited in breeding programmes aimed at producing genotypes with higher K efficiency. Particular attention will be paid to putative mechanisms underlying K efficiency.

Soil K pools

Soil K can be assigned to four distinct pools differing in availability to plants: (1) soil solution K, (2) exchangeable K, (3) non-exchangeable K (positioned in interlayers of clay minerals, especially those of the 2:1:1 type) and (4) structural K (Moody and Bell 2006). Plants take up K exclusively from the soil solution pool, which is in a dynamic equilibrium with the exchangeable and, to a lesser extent, the non-exchangeable pools. Exchangeable K can be rapidly released from exchange sites on the surfaces of clay minerals and organic matter to replenish K-depleted soil solution (Steingrobe and Claassen 2000). Non-exchangeable K can also be released into soil solution, with the data for a Luvisol showing such release being triggered when soil solution K^+ concentration dropped below 3.5 μM (Springob and Richter 1998b). However, a release of non-exchangeable K from interlayer sites of clay minerals is a slow process and is mostly important in contributing to the replenishment of the soil solution and exchangeable pools in the long-term (e.g. over successive crops; Pal et al. 2001a, 2002). The release of structural K into soil solution can be effected only by weathering of clay minerals; hence, it is an exceedingly slow process with no discernible effect during a single crop cycle (Pal et al. 2001b).

Species and genotypes differ in their capacity to mobilize non-exchangeable K. In contrast to wheat and potato, sugar beet can facilitate the release of K from the non-exchangeable pool into the rhizosphere soil solution (Steingrobe and Claassen 2000). Potato cultivars that are K-efficient have a capacity to chemically mobilize non-exchangeable K within days (Trehan and Sharma 2002, Trehan et al. 2005). Hence, in soils with a significant clay fraction and therefore a potentially extensive non-exchangeable K pool, the capacity of a genotype to influence the dynamics of K release from that pool to the soil solution can influence efficiency of K uptake (El-Dessougi et al. 2002, Trehan 2005).

Potassium efficiency: what is it?

The capacity of a genotype to grow and yield well in soils low in K availability is termed K efficiency. The K efficiency was described as the proportion of yield potential that can be achieved under K deficiency stress (Damon and Rengel 2007, Damon et al. 2007). K-efficient genotypes can have higher acquisition of K from soil (uptake efficiency) and/or higher dry matter production per unit of K taken up (utilization efficiency). The K uptake and utilization efficiency are interlinked. Crop genotypes with the capacity to take up relatively more K where availability was low (Chen and Gabelman 1995, Trehan and Sharma 2002, Zhang et al. 2007) have been identified together with those that produce a relatively large biomass per unit of K taken up where K availability in the soil is low (El-Bassam 1998; Woodend et al. 1987, Yang et al. 2003). A positive correlation was found between K utilization efficiency at low available K and relative shoot dry weight (shoot dry weight at deficient/adequate K supply) at the tillering stage (Yang et al. 2003). For sweet potato, K utilization efficiency was positively correlated with total plant biomass and root yield (George et al. 2002).

In terms of the sustainable use of native soil K resources, particularly in sandy soils where the total K pool is small, the amount of K removed in the economic product is highly relevant. However, low K concentration in the economic product is generally not a desirable trait because of the reduced nutritional value for human consumption. Nevertheless, for some production systems (e.g. maize; Muhr et al. 1999), it may be beneficial to identify genotypes that remove less K in the exported product, whereas in other crops, such as wheat, the amount of K in the exported product is low and is of less importance to the sustainability of the farming enterprise. For high input systems such as cotton (Brouder 1994 #8693), it may be desirable to select genotypes that obtain maximum yield response from applied K (Brouder and Cassman 1990).

Genotypic differences in K efficiency

Considerable variation in efficiency of K uptake and utilization has been identified among existing genotypes for a variety of crop species, including lucerne (James et al. 1995), beans (Shea et al. 1968), soybean (Sale and Campbell 1987), tomato (Chen and Gabelman 1995, Figdore et al. 1989), ramie (*Boehmeria nivea*) (Liu et al. 2000), maize (Farina et al. 1983), wheat (Damon and Rengel 2007, Woodend et al. 1987), barley (Pettersson and Jensén 1983), rice (Yang et al. 2004), potato (Trehan et al. 2005, Trehan and Sharma 2002), sweet potato (George et al. 2002), canola (Damon et al. 2007), cotton (Zhang et al. 2007), etc. (Table 1).

Significant variation in K utilization efficiency (as the amount of biomass or economic yield per unit of K taken up) has been reported among genotypes for a number of crop species, including wheat (Damon and Rengel 2007, El-Bassam 1998, Glass and Perley 1980, Woodend and Glass 1993, Zhang et al. 1999), rice (Yang et al. 2004), barley (Shivay et al. 2003), sweet potato (George et al. 2002) and canola (Damon et al. 2007) (Table 1). It is reasonable to assume that genotypic variation in K utilization efficiency exists within all the major crop species. Although numerous studies have identified inter- and intraspecies differences in K efficiency, relatively few have made detailed assessments of the observed differences at a mechanistic level and none has taken a strategic approach towards breeding selection, inheritance or molecular markers for key K efficiency mechanisms.

Mechanisms of K efficiency

There is a strong relationship between uptake and utilization efficiency in producing K-efficient phenotype under given set of conditions (Fig. 1). Potassium uptake efficiency is governed by mechanisms relying on root architecture, the high uptake capacity at the root surface as well as the capacity to mobilize non-exchangeable K by root exudates. Potassium utilization efficiency is based on effective K translocation between organelles, cells and organs, the capacity to use other ions as substitutes for K and the capacity to direct assimilates to the harvested parts.

Potassium uptake efficiency

Uptake efficiency of soil-grown plants may consist of increased capacity to mobilize/solubilize non-available nutrient forms into plant-available ones, and/or increased capacity to transport nutrients across the plasma membrane of root cells and into other plant parts. For K, it appears likely that the increased capacity to convert non-available into available forms is of greater importance for efficient uptake because K is transported to roots by diffusion, which is a relatively slow process that frequently limits nutrient uptake. However, larger amounts are transported towards roots if a larger concentration gradient between the root surface and the bulk soil can be maintained by vigorous K uptake at the root surface (Jungk and Claassen 1997). When a capacity of root cells to take up K exceeds the rate of K replenishment at the root surface, the uptake rate will be governed by the K supply rather than by the capacity of plants to take K up (cf. Rengel 1993). Therefore, an increased capacity of root cells to take up K because of, for example, an increased expression of high-affinity K uptake systems in the plasma membranes under nutrient-deficiency stress is expected to be of secondary importance as an efficiency mechanism for K and other diffusion-supplied nutrients (e.g. P, Zn, Mn and Cu).

Root morphology

For diffusion-supplied nutrients (like K), it is important to have a large surface area of contact between roots and soil, which is the result of having thin roots. Hence, K-efficient genotypes could have a relatively larger proportion of thin roots in the whole root system compared with K-inefficient genotypes. However, no study to confirm such a hypothesis could be found. However, the hypothesis has been well supported in terms of Zn-efficient and Zn-inefficient genotypes (Dong et al. 1995, Rengel and Wheal 1997). On the other side, fast root turnover may be important for K efficiency of several plant species upon exposure to low K availability (Steingrobe 2005).

In some sandy soils, leaching of K may be a significant loss of K from the system, and it may be beneficial to introduce crops and genotypes that grow roots and take up more K from deep in the soil profile and transport it to aboveground plant parts (see Wong et al. 2000), so that K is more sustainably cycled through the soil-crop-stubble continuum.

Plants may not have the capacity to opportunistically increase root growth in the areas of high K availability; nevertheless, K^+-rich patches induced a global response and accelerated the growth of laterals even in those root segments in barley that were exposed to low K concentrations (see Ashley et al. 2006). However, genotypic K efficiency may not be linked to increased root growth; in contrast, K-efficient potato genotype had half the root length of the K-inefficient cultivars, despite having similar relative shoot growth rate, but K-efficient genotypes had higher K influx than K-inefficient ones (Trehan and Sharma 2002).

Root hairs

Root hairs contribute to the K uptake capacity by increasing the surface area and the radius of the K

Table 1. Compilation of literature on differential K efficiency in economically important plant species.

Crop species	Genotypes tested	Culture medium	Environment	K rates applied (as K)	Growth stage of assessment	Suggested mechanisms for differences in K efficiency	Reference
Wheat (*Triticum aestivum* L.)	154	Sandy soil	Glasshouse	0 or 88 mg kg^{-1}	5 weeks	Harvest index, shoot yield per unit of K taken up	Damon and Rengel (2007)
	12	Sandy soil	Glasshouse	15 or 100 mg kg^{-1}	Zadok 31 and maturity		
	88	Sandy soil	Field	0 or 100 kg ha^{-1}	Zadok 4 and maturity		
	58	Soil	Field	Nil K only	Maturity	Shoot and grain yield per unit of K taken up, harvest index	Zhang et al. (1999)
	4	Soil	Field	0 or 111 kg ha^{-1}	Maturity		
	3	Nutrient solution	Glasshouse	0–10 m*M*	Tillering		
	21	Loamy sand	Field	83 kg ha^{-1}	Maturity	Shoot and grain yield per unit of K	El-Bassam (1998)
	16	Nutrient solution	Growth room and natural conditions	10 μ*M* K	3 weeks and maturity	Harvest index, shoot/grain yield per unit of K taken up	Woodend et al. (1987)
Rice (*Oryza sativa* L.)	134	Nutrient solution	Glasshouse	5 or 40 mg l^{-1}	Maturity	Shoot and grain yield per unit of K taken up, harvest index	Yang et al. (2003)
	9	Soil	Field	0 or 78 kg ha^{-1}	Tillering and maturity	Shoot and grain yield per unit of K taken up, harvest index	
	9	Silty loam soil	Field	157 kg ha^{-1}	Maturity	Harvest index, shoot/grain yield per unit of K taken up, translocation of K to upper leaves	Yang et al. (2004)
Barley (*Hordeum vulgare* L.)	10	Nutrient solution	Growth room	0.1–500 μ*M*	2 weeks	K influx rate	Glass and Perley (1980)
	3	Nutrient solution	Growth room	Nil	Young seedling	Subcellular distribution of K	Memon et al. (1985)
	11	Nutrient solution	Growth room	3 m*M*	Young seedling	Shoot yield per unit of K taken up	Jensen and Pettersson (1980)
Canola (*Brassica napus* L.)	84	Sandy soil	Glasshouse	0 or 88 mg kg^{-1}	37 days old	K uptake, shoot yield per unit K absorbed	Damon et al. (2007)
	4	Sandy soil	Glasshouse	0 or 125 mg kg^{-1}	54 days and maturity	No differences	Brennan and Bolland (2007)
Potato (*Solanum tuberosum*)	10	Luvisol subsoil	Natural conditions	0 or 195 mg kg^{-1}	29 and 83 days	Uptake of non-exchangeable K, root length/shoot ratio	Trehan et al. (2005)
	3	Sandy loam/sand	Natural conditions	0 or 100 mg kg^{-1}	19 and 45 days	Uptake of non-exchangeable K, K influx rate	Trehan and Sharma (2002)
Common bean (*Phaseolus vulgaris* L.)	10	Oxisol	Glasshouse	0 or 200 mg kg^{-1}	Maturity	K partitioning, harvest index	Fageria et al. (2001)
Tomato (*Lycopersicon esculentum* Mill.)	4	Sand/zeolite	Growth room	nil K only	3 weeks	Root length, high K influx	Chen and Gabelman (2000)
	5	Nutrient solution	Growth room	5 mg pot^{-1}	30–44 days old	Na/K substitution capacity	Figdore et al. (1989)
Sweet potato (*Ipomoea batatas* L.)	8	Sandy loam Soil	Field	0 or 112 kg ha^{-1}	150 days	Root and total biomass yield per unit of K taken up	George et al. (2002)
Lucerne (*Medicago sativa* L.)	9	Soil	Glasshouse	0, 100 or 200 mg kg^{-1}	Early bloom	Shoot yield per unit of K taken up	James et al. (1995)

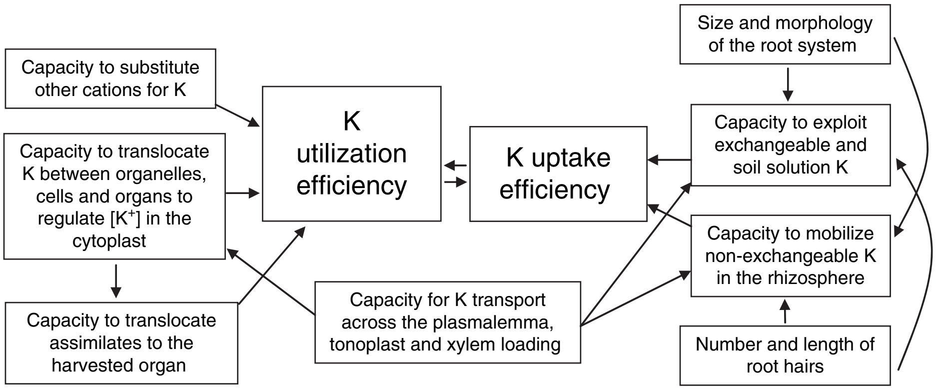

Fig. 1. Potential mechanisms of K uptake and utilization efficiency and their interactions that influence the expression of the K-efficient phenotype.

depletion zone. Root hairs create a steeper diffusion gradient simply because they are thinner than roots (Jungk 2001). In pot experiments with maize, oilseed rape, tomato, ryegrass and onion, the radial distance of the K (Rb) depletion zone adjacent to the root surface increased with the length of the root hairs (see Jungk 2001). The capacity to take up K from a K-deficient soil was positively correlated with root hair length for pea, red clover, lucerne, barley, rye, perennial ryegrass and oilseed rape (Hogh-Jensen and Pedersen 2003). All crops increased root surface area at low K supply several-fold by increasing root hair length, but the amount or density of root hairs did not change. Although all species exploited the exchangeable K sources similarly, the crop species with the highest root surface exploited the most sparingly soluble K.

Although root hair formation is known to differ between genotypes of the same species (Jungk 2001), no literature could be found which assessed the formation of root hairs as a mechanism for intraspecific differences in efficiency for K uptake. It is also somewhat unfortunate that none of the studies applying model predictions have measured or incorporated the effect of root hairs. In studies where root hairs were measured (Claassen and Jungk 1984; Hogh-Jensen and Pedersen 2003), they were indeed found to be a major factor in K uptake.

Root exudates

Sugar beet took 3–6 times more K per unit of root length than wheat and barley grown in a K-fixing soil. Model calculations predicted that the K concentration difference between the surface of sugar beet roots and the bulk soil was up to seven times higher than the measured K concentration in the soil solution, indicating that high capacity to mobilize K in the rhizosphere may be the mechanism for the observed efficiency for K uptake by sugar beet (El-Dessougi et al. 2002).

Enhanced mobilization of mineral K might be attributed to the release of organic acids from the plant roots because exposure of gneiss to malic and tartaric acids resulted in the release of mineral K from gneiss (Wang et al. 2000). Similarly, the amino acids detected in root exudates of wheat and sugar beet were found to desorb K from a K-fixing soil, but could not account for the K-uptake efficiency observed in sugar beet (Samal 2007). Likewise, exudates from oat roots increased the K availability in a Luvisol subsoil, albeit only slightly (Liebersbach et al. 2004). Differential root exudation could be at least partly responsible for differential K efficiency of different genotypes (e.g. potato, see the section below; Trehan et al. 2005).

Release of K from the non-exchangeable pool

Model predictions generally indicate that diffusion is a minor factor in K efficiency, and the capacity to utilize non-exchangeable K is more important (Claassen and Steingrobe 1999, Claassen et al. 1986). Indeed, differential K efficiency of 10 potato cultivars was largely attributable to their differential uptake of non-exchangeable K from soil (Trehan et al. 2005). K-efficient genotypes of potato obtained 46% of K from the non-exchangeable pool, whereas the K-inefficient genotypes achieved only 17–25% in a glasshouse study (Trehan and Sharma 2002). Nine of 10 cultivars took up 1.6–2.8 times more K than was extracted from soil by ammonium acetate. Calculated K concentration at the root surface was similar among genotypes; hence, the concentration gradient towards the root surface was probably not the principal driver of K diffusion and subsequent uptake. It was proposed that the observed genotypic differences in the uptake of non-exchangeable K were because of exudation of K-mobilizing compounds from roots (Trehan et al. 2005). For example, comparison of experimental data with model predictions indicated that wheat was K-efficient because of a high root:shoot ratio, whereas sugar beet was able to increase the availability of K in the rhizosphere (Steingrobe and Claassen 2000).

Inoculation with bacterial strains capable of releasing K from non-exchangeable pools may increase K uptake (Supanjani et al. 2006). *Bacillus edaphicus* can release K from illite (Sheng 2005, Sheng and He 2006) and feldspar (Sheng and He 2006). This bacterium can get established in the rhizosphere of cotton and rape (Sheng 2005) as well as wheat plants (Sheng and He 2006), contributing to increased K uptake by these plants. Positive effects of other bacteria on K release from water-insoluble K rocks have also been reported, e.g. *Bacillus mucilaginous* improved maize growth and K uptake (Wu et al. 2005). However, no report could be found where differential capacity of plant genotypes to support K-releasing bacteria in the rhizosphere or take up bacterially released K was documented.

Kinetics of K uptake

The capacity of roots to take up K^+ at a higher rate at low external concentrations has a significant potential to influence efficiency of K uptake. According to widely accepted Michaelis-Menten kinetics, C_{min} represents the minimum external K concentration at which K is taken up. Because K uptake in soils is largely diffusion limited, C_{min} can influence not only K uptake at the root surface, but also the rate of K delivery to the root surface via diffusion. Because depletion of K^+ in soil solution can initiate the release of non-exchangeable (interlayer) K (Springob and Richter 1998a, 1998b), greater uptake capacity of the root cells via high-affinity K uptake systems may potentially increase the availability of K in the rhizosphere (X Wei, P Damon, O Babourina and Z Rengel, unpublished data). Indeed, in the nutrient solution, wheat genotypes differed in K uptake parameters (V_{max} and K_m) under low K^+, suggesting differences in the high-affinity K uptake system (Glass and Perley 1980). In soil, however, the difference between wheat genotypes was expressed as higher K uptake per unit of root length in K-efficient compared with K-inefficient genotype only at high K fertilization (Fig. 2).

The rate of K uptake is an important mechanism of K uptake efficiency in some species. For example, potato required nine times higher external K concentration than wheat or sugar beet to achieve 90% of maximum yield in flowing nutrient solution culture. This could not be accounted for by the difference in internal K requirement, which was only 1.5 times higher, but instead sug-

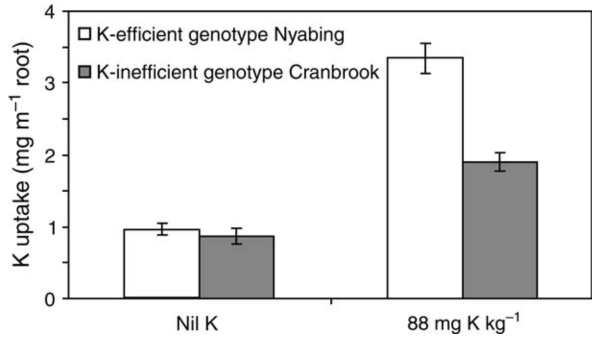

Fig. 2. Average K uptake by wheat genotypes growing for 8 weeks in 1-m deep columns of K-deficient soil without or with K applied at 88 mg kg^{-1} soil. Means ± SE (PM Damon and Z. Rengel, unpublished data).

gested differences in the high-affinity K uptake mechanism (Trehan and Claassen 1998). In contrast, differences in the low-affinity K^+ uptake mechanism among vine rootstocks (Ruhl 2000) and wheat genotypes grown in solution (Glass and Perley 1980) were suggested to be in a core of differential K efficiency.

In potato grown at K deficiency, K-efficient genotype had about two-fold higher K uptake rate than K-inefficient one (Trehan and Sharma 2002). In tomato, high K uptake was identified as the mechanism for 18 of 22 tomato genotypes classified as K efficient, with root length proliferation singled out as the main mechanism for the other four genotypes, although root hairs were not assessed (Chen and Gabelman 1995). Subsequent work (Chen and Gabelman 2000) confirmed K efficiency in tomato being because of high net K uptake coupled with low pH at the root surface. However, the high K uptake was attributed to mobilization of K in the rhizosphere by root exudates, creating a high K concentration in soil solution and apoplast, rather than having high capacity for K uptake at the root surface.

K utilization efficiency

Genotypic differences in capacity to utilize potassium can be attributed to differences in the (1) partitioning and redistribution of K at a cellular or whole-plant level, (2) substitution of K with other ions, and/or (3) the partitioning of resources into the economic product (Gerloff 1987, Sattelmacher et al. 1994). For optimal metabolic activity, plants maintain a K concentration in the range of 60–150 m*M* in the cytosol (Amtmann et al. 2005, Cuin et al. 2003, Karley et al. 2000, Leigh 2001), although vacuolar K^+ concentrations may vary between 10 and 200 m*M* and even up to 500 m*M* in the guard cells (Ashley et al. 2006) and cytosolic K^+ concentration may be influenced by external K^+ and NH_4^+ concentrations (Kronzucker et al. 2003). Potassium concentration in the cytosol and the vacuole varies with the cell type as well as plant age (Fricke et al. 1994). Differences in the K requirements of tissues are generally attributable to differences in the K requirements in the vacuoles and apoplast as well as the capacity to maintain cytosolic K at a certain level.

Efficient utilization of K during the vegetative stage may not translate into K efficiency for economic yield. The genotypic differences in harvest index influenced the efficiency with which K could be utilized to yield grain more than the capacity to utilize K to produce biomass (e.g. in wheat; Woodend and Glass 1993). Another study with wheat found that K utilization efficiency at stem elongation was positively correlated with shoot weight and grain yield at maturity, but was not correlated with harvest index (Damon and Rengel 2007).

Translocation

Potassium is highly mobile in plants and differences between genotypes in efficiency of K utilization have been attributed to differences in capacity to translocate K at a cellular and whole plant level. Under K^+ deficiency, cytosolic K^+ activity is maintained at the expense of vacuolar K^+ activity (Leigh 2001, Memon et al. 1985), even though vacuolar (but not cytosolic) K^+ activity is regulated differently in the root and leaf cells (Cuin et al. 2003). Mobilization of vacuolar K into the cytosol was particularly strong in K-efficient compared with K-inefficient barley genotypes (Memon et al. 1985).

Capacity to translocate K between organs may also be an important mechanism for efficient utilization of K within the plant (in ryegrass; Dunlop and Tomkins 1976). Capacity to translocate K from non-photosynthetic organs such as stems and petioles to upper leaves and harvested organs can influence the genotypic capacity to produce a high economic yield per unit of K taken up. Two K-efficient rice genotypes had two-fold higher K concentration in lower leaves, but only 30% higher K concentration in the upper leaves compared with K-inefficient rice genotype at the booting stage (Yang et al. 2004). The two K-efficient genotypes also had higher stomatal conductance, net photosynthetic rate and RuBPcase activity in upper and (particularly) lower leaves than for a K-inefficient genotype. By maintaining a higher K concentration in the lower leaves, the K-efficient genotype was able to maintain a larger photosynthetic capacity during grain filling.

The high-yielding bean cultivar Diamante Negro was K efficient in terms of grain yield per unit of K taken up and had the lowest proportion of aboveground K allocated to grain (Fageria et al. 2001). By having an apparently low requirement for K in grain, this genotype was able to produce high yield at low K supply, even though harvest index was not high.

Substitution

Accumulation of K in vacuoles creates the necessary osmotic potential for cell extension. Rapid cell extension requires high mobility of the osmoticum, so only few other ions (e.g. Na^+) can replace K in this role. Osmotic maintenance can then be carried out by less mobile molecules such as sugars, and K ions can be partially replaced and recovered from vacuoles (Amtmann et al. 2005). The Na^+ ions can also substitute for K^+ in enzymatic reaction leading to organic acid production (Ehrendorfer 1973). The potential for Na^+ as a partial substitute for K^+ in some, but not all plant species, is well documented (Marschner 1995).

Tomato genotypes were identified with high-, intermediate- or low-substitution capacity for Na^+, with regard to their growth response to Na^+ application under condition of deficient K supply. Two genotypes had an average increase in plant dry weight of 50% over the respective non-salt control treatment, while one genotype had only a 6% increase in dry weight over the minus Na control (Figdore et al. 1989), clearly indicating genotypic differences in the capacity to use Na^+ to substitute for K^+ in physiological functions. It is also worth noting that the salt-tolerant wild relative of tomato (i.e. *Lycopersicon pennellii*) has more efficient K^+ substitution by Na^+ than the cultivated tomato, accompanying with increased K utilization efficiency (Taha et al. 2000).

Harvest index

As the proportion of total plant weight that is economic yield, harvest index quantifies the capacity of a genotype to allocate resources, including K, to the harvested organs. A high harvest index is fundamental to efficient utilization of all resources taken up by the plant and is therefore of significant interest to breeders. The dramatic yield improvements for many crops that have occurred during the 20th century have been largely because of improvements in harvest index (Hay 1995). For example, whilst modern varieties of winter wheat were found to produce on average 60% more grain than the old ones, the increase in total shoot biomass was only 6% (see Dambroth and El-Bassam 1983).

The importance of high harvest index as a mechanism of K utilization efficiency has been widely documented for a number of species including wheat (Damon and Rengel 2007, Woodend and Glass 1993, Zhang et al. 1999) (see Fig. 3), rice (Yang et al. 2004), sweet potato (George et al. 2002) and common bean (Fageria et al.

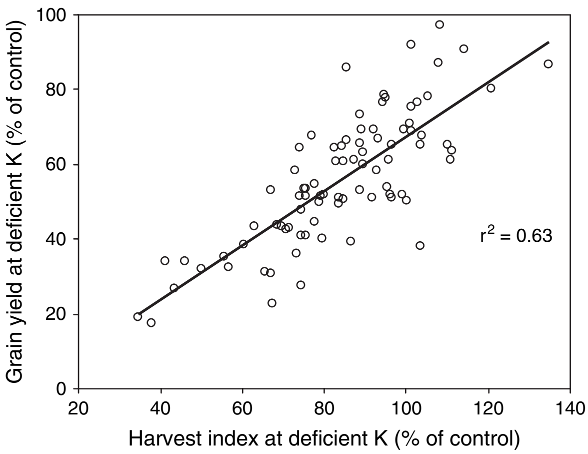

Fig. 3. Response of harvest index and grain yield to K deficiency stress for 85 genotypes of wheat grown to maturity in the field (modified from Damon and Rengel, 2007).

2001). High harvest index at deficient K supply and the high ratio of harvest index at deficient to adequate K supply were the main factors determining tolerance to K deficiency in terms of grain yield for wheat genotypes in the field (89 genotypes) and in the glasshouse (12 genotypes) (Damon and Rengel 2007).

Harvest index has such a dominant influence on K efficiency for some species so that vegetative measures for K efficiency can be poorly correlated with K efficiency for grain yield. Indeed, the vegetative measures of K efficiency were unreliable indicators of the K efficiency for the economic product if the genotypes differed significantly in their harvest index (e.g. in 16 genotypes of wheat; Woodend and Glass 1993). However, for other species, the magnitude of genotypic differences in harvest index had a less profound effect on K efficiency. The grain yield for 10 bean genotypes under K deficiency stress was predominantly determined by shoot biomass and, to a lesser extent, harvest index (Fageria et al. 2001).

Potassium utilization efficiency of sweet potato genotypes was positively related to harvest index (George et al. 2002). Similarly in rice, K-efficient rice genotypes had higher harvest indices than inefficient ones when grown at low K (Yang et al. 2004). Harvest index was positively correlated with both grain yield and biomass per unit of K taken up. Grain filling rate was seven times lower during the second week of grain filling (but similar in the first week) for a K-inefficient rice genotype compared with two K-efficient rice genotypes in a K-deficient soil (Yang et al. 2004). For four genotypes of wheat grown in the field, grain produced per unit of K taken up into aboveground parts was positively correlated with harvest index, and negatively correlated with stem K concentration at maturity (Zhang et al. 1999) (see also Fig. 3), indicating that efficient utilization of K within the plant was attributable to capacity to allocate biomass to grain, as well as a capacity to tolerate low concentration of K in stems.

Breeding

Breeding of nutrient-efficient genotypes is expected to progress through stepwise compounding of genetic information for a number of nutrient efficiency mechanisms into locally adapted crop cultivars (Rengel 2005). In such a breeding programme, genotypes with genes that control a particular mechanism of efficiency may be important, even though they themselves may not show phenotypically high nutrient efficiency.

One of the major problems with nutrient deficiency is that the level of stress is frequently assessed based on plant growth. However, growth, as a very complex character influenced by a myriad of inter-related processes, is clearly not a suitable parameter for distinguishing causes and effects of nutrient deficiency stress. Better screening techniques need to be developed by targeting specific processes that cause (or prevent) nutrient deficiency, rather than those that appear as a consequence of it (Rengel 2005).

So far, there has been slow and fragmented progress towards developing K-efficient cultivars suitable for incorporation into production systems, even though heritability of K efficiency is high (at least in lucerne; James et al. 1995) and selection for K utilization efficiency may be effective across field environments, e.g. in *Lolium perenne* (Smith et al. 1999). Critical steps for developing K-efficient genotypes are: (1) identifying differences in K efficiency in the germplasm of agricultural species (Damon and Rengel 2007, Damon et al. 2007, Stelling et al. 1996, Trehan et al. 2005), (2) elucidating the principal mechanisms driving the observed differences in K efficiency (Damon and Rengel 2007, El-Dessougi et al. 2002, Figdore et al. 1989) and (3) identifying suitable molecular markers or screening procedures by which the desirable traits can be efficiently recognized within large breeding programmes.

Quantitative trait loci (QTLs) for K efficiency have been reported only in rice (Wu et al. 1998). However, additional QTLs have been found for shoot K concentration in rice (Lin et al. 2004) and Arabidopsis (Harada and Leigh 2006). It is expected that in the near future molecular methods will dominate selection for nutrient efficiency. Development of molecular markers for regions of the genome segregating with the trait of K efficiency will be crucial in that respect. However, molecular markers may be population dependent (e.g. markers for Fe efficiency in soybeans, Cianzio 1999), making them unsuitable for use in breeding programs.

Future work

Inclusion of K-efficient genotypes into farming systems will increase efficiency of K cycling in the soil-crop-stubble continuum and will be particularly important in low-input agriculture. Breeding K-efficient genotypes will benefit greatly from better understanding of the relevant efficiency mechanisms as well as molecular markers associated with such mechanisms to allow marker-assisted selection. Complementary agricultural management practices will be required for genotypes with increased K efficiency to allow for full economic benefit of improved germplasm.

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